

Quantitative Impact of Global and Domestic Events on the BIST

Project Final Report *DSA210* - Yiğit Keskin

1. MOTIVATION

Emerging financial markets, particularly the Borsa Istanbul (BIST 100), operate within high-volatility microstructures characterized by acute sensitivity to shifting macro-environments. Volatility and asset return anomalies do not develop inside a vacuum; they are tightly bound to sudden exogenous shocks (such as geopolitical altercations, natural disasters, or global healthcare emergencies) and structural endogenous pivots (such as sudden adjustments to interest rates or currency interventions by the central bank).

This project investigates how major domestic and global turning points quantitatively shape the BIST 100 index from 2015 to the present. Rather than analyzing asset movements through simple isolated trendlines, this study applies a thorough data science pipeline to evaluate whether external macroeconomic crises structurally deform market return distributions and risk variance. It further tests the predictive boundaries of standard machine learning paradigms to observe whether complex models can accurately forecast directional tail-risk or out-of-sample asset returns under systemic stress.

2. RESEARCH QUESTIONS AND SUB-QUESTIONS

Main Question

- To what extent do global exogenous shocks and domestic macroeconomic policy shifts drive return distribution and volatility expansion within the BIST 100 index?

Sub-Questions

1. Does the depth of a market drop correlate directly with the assigned qualitative severity level of an external shock, or do market participants react indiscriminately once a structural crisis hits?
2. Does the BIST 100 exhibit a significant short-term "hangover" effect where market panic spills over into subsequent trading sessions, or does it operate with near-immediate informational efficiency?

3. Can complex machine learning classification and regularized regression pipelines accurately forecast future daily returns using time-lagged historical indicators under crisis regimes?

3. HYPOTHESES

- **H1:** The distribution of daily returns on macro-shock days is fundamentally shifted and statistically different from normal trading days.
- **H2:** Macroeconomic and geopolitical shocks systematically trigger a massive expansion of market volatility (variance) compared to standard trading conditions.
- **H3:** The magnitude of the immediate market drop depends strictly on the categorized qualitative severity level (Medium vs. High vs. Extreme) of the external shock.
- **H4:** The market exhibits a short-term "hangover" effect, causing negative pressure on returns to persist into the trading session immediately following an event (T+1).
- **H0 (Null Hypothesis):** External historical shocks and macroeconomic indicators hold no statistically significant or predictive relationship with BIST 100 return distributions or volatility parameters.

4. DATA DESCRIPTION

Core Data Dictionary

The quantitative panel consists of three foundational layers extracted from raw environments and compiled into unified datasets:

- **Yahoo Finance (bist100_raw.csv):** Daily market data tracking Open, High, Low, Close, and Volume parameters. Serves as the independent target and baseline asset pricing matrix.
- **TCMB EVDS (macro_indicators_raw.csv):** Houses monthly Consumer Price Index (CPI) metrics (TP.FG.J0) and daily USD/TRY Exchange Rates (TP.DK.USD.A.YTL) acting as fundamental economic context layers.
- **Event Catalog (events_raw.csv):** A customized master log detailing 24 significant exogenous and endogenous shock dates between 2015 and 2026, filtered by thematic category and baseline severity thresholds.

Pipeline Dataset Generation

The data acquisition scripts clean, align, and output three primary analytical panels:

1. **unified_bist_data.csv:** Merged daily time-series with explicit binary event flags used for baseline exploratory data visualizations.
2. **shock_analysis_summary.csv:** An event-level window panel isolating pre- and post-event returns for formal statistical hypothesis processing.

3. **unified_bist_enriched.csv**: Enriched daily master panel incorporating rolling technical metrics and time-lagged features engineered specifically for downstream machine learning validation loops.

5. DATA SOURCE AND COLLECTION

Financial asset price configurations were dynamically acquired via the open-source yfinance API wrapper using the target index symbol XU100.IS. The dataset captures a continuous historical timeline from January 2, 2015, to April 3, 2026, comprising 2,817 active trading observations.

Macroeconomic indicators were derived from programmatic integration with the Central Bank of the Republic of Türkiye (TCMB) Electronic Data Delivery System using the evdsAPI Python client library. The qualitative historical event logs were cross-referenced against public economic news timelines to define severe turning points (including the 2016 domestic coup attempt, the 2018 currency stress crisis, the 2020 global pandemic declaration, and the 2023 natural disasters).

To prevent look-ahead bias and structural alignment errors, all monthly inflation disclosures were forward-filled (ffill) across the daily trading calendar, creating an uncorrupted daily time-series panel.

6. METHODOLOGY

6.1 Panel Alignment & Technical Feature Engineering

Because macroeconomic shocks frequently occur outside standard equity trading boundaries (e.g., weekends, institutional bank holidays), an asymmetric, forward-looking look-up mechanism was deployed using pandas merge_asof with a direction set to forward. This architecture automatically aligns a weekend or holiday crisis date directly onto the market opening bell of the first available trading session.

To capture momentum shifts and asset price elasticity away from baseline paths, three technical indicators were engineered across the unified data panel:

- **7-Day Rolling Volatility**: Evaluated via a moving standard deviation of daily return margins to record localized baseline risk expansion.
- **20-Day Moving Average (MA)**: Charts the persistent mid-term pricing trend of the index.
- **Trend Deviation Indicator (dev_from_ma)**: Measures the proportional percentage "snap" distance from the active moving average to flag severe market oversold or overbought boundary breaks during panic cascades. Calculated as: (Close minus 20-Day Moving Average) divided by the 20-Day Moving Average.

6.2 Exploratory Data Architecture

Visual profiling was deployed across data layers using matplotlib and seaborn. Independent Kernel Density Estimation (KDE) overlays were constructed to visibly contrast normal return profiles against shock boundaries, illustrating non-Gaussian fat-tail properties. To bypass local currency inflation skewing long-term capital representations, a nominal conversion process was embedded to track the BIST 100 priced strictly in USD terms (close divided by `usd_try`), recording true macroeconomic wealth preservation trends.

6.3 Inferential Statistical Testing

Financial time-series data naturally violates standard parametric assumptions due to persistent skewness, high kurtosis, and non-linear outliers. Therefore, standard t-testing methodologies were bypassed in favor of robust alternatives:

- **Mann-Whitney U Test (H1 & H4):** A two-sided non-parametric evaluation comparing localized sample medians. This architecture tests if return distribution midpoints fundamentally shift under stress without assuming an underlying normal curve layout.
- **Levene's Test for Homogeneity of Variance (H2):** Deployed to mathematically evaluate volatility expansions across normal vs. shock groups due to its inherent structural resistance to non-normal return distributions.
- **One-Way ANOVA Regression (H3):** Estimated via ordinary least squares (OLS) formulations to verify whether qualitative severity designations explain variance changes in continuous market return depths.

6.4 Predictive Machine Learning Framework

A strict out-of-sample validation matrix was implemented within scikit-learn to analyze the bounds of predictive modeling. Independent variables were shifted backward by exactly one period (T-1) to guarantee that predictive tasks are executed using exclusively historical information, eliminating data leakage.

Data partitions were constructed via a time-aware chronological split (First 50% for model fitting, final 50% held out for out-of-sample evaluation), maintaining true chronological execution boundaries. Because market shock anomalies are rare (N=18 instances preserved within the localized model splits), Synthetic Minority Over-sampling (SMOTE) parameters were dynamically embedded inside an encapsulated pipeline wrapper to prevent structural classifier collapse without contaminating out-of-sample evaluation spaces.

7. RESULTS AND FINDINGS

7.1 Inferential Statistical Results

The empirical evaluation demonstrates that while structural macroeconomic events heavily dictate systematic risk levels, the index displays advanced informational absorption capabilities.

Statistical Test Suite Summary

- **Hypothesis 1 (Return Shift):** Tested via Mann-Whitney U Test. Output: $U = 17883.0$. Result: Supported (P-Value: $3.3786e-02$). *Academic Insight:* Macro-shocks significantly alter return medians. Shock days skew heavily negative compared to standard baseline trading sessions.
- **Hypothesis 2 (Volatility Expansion):** Tested via Levene's Variance Test. Output: Levene Stat = 17.5025. Result: Supported (P-Value: $2.9570e-05$). *Academic Insight:* Shocks trigger massive system-wide risk expansion. Shock day return volatility measures 1.91x higher than normal trading conditions.
- **Hypothesis 3 (Severity Sorting):** Tested via One-Way ANOVA Regression. Output: F-Statistic = 0.5840. Result: Not Supported (P-Value: 0.5698). *Academic Insight:* Once a crisis strikes, investor panic behaves indiscriminately. A "Medium" event triggers asset liquidations just as severe as an "Extreme" designation.
- **Hypothesis 4 (Hangover Effect):** Tested via Lagged Mann-Whitney U. Output: T+1 Return Shift. Result: Not Supported (P-Value: $6.9933e-01$). *Academic Insight:* The BIST 100 operates with high efficiency. Shock vectors are absorbed within a single trading day; no risk spillover carries into Day T+1.

7.2 Machine Learning Performance Analysis

Predictive tasks executed across the out-of-sample data partitions exposed stark differences between predicting absolute returns and classifying systemic risk environments.

Continuous Regression Models (Absolute Return Estimation)

Forecasting exact daily asset return percentages using lagged variables proved extremely difficult, aligning with the core tenets of the Efficient Market Hypothesis (EMH).

- **Linear Regression Performance:** R-Squared = -1.404 (RMSE = 2.704%)
- **Random Forest Regressor Performance:** R-Squared = -0.071 (RMSE = 1.805%)

Both standard linear formulations and complex non-linear tree ensembles generated negative out-of-sample R-Squared parameters. This mathematical reality proves that complex model parameters overfit cross-sectional market noise under high-stress regimes, completely failing to outperform a simple naive baseline tracking the historical index mean.

Categorical Risk Classification (Risk Detection)

While continuous return level forecasting hit major structural limits, mapping model boundaries onto categorical risk identification provided practical risk-management application.

- **Baseline Logistic Regression Framework:** Generated an out-of-sample Area Under the Curve (ROC-AUC) metric of 0.392 when evaluated under a custom threshold parameter of 0.40.
- **SMOTE-Optimized Random Forest Pipeline:** Significantly expanded predictive capabilities, generating a superior out-of-sample ROC-AUC parameter of 0.512.

Permutation importance algorithms executed across the out-of-sample ROC-AUC partitions verified that 7-day lagged rolling volatility holds the primary explanatory power for predictive tasks, heavily outranking lower-frequency variables like the monthly CPI inflation tracking layer. This confirms that while machine learning pipelines cannot reliably forecast daily directional arbitrage returns during emerging market crises, non-linear ensemble models can isolate high-stress environments to activate systematic early warning alerts.

7.3 Main Academic Analytical Contributions

By bridging advanced financial theory with predictive data engineering, this investigation establishes three primary analytical contributions to the study of emerging market microstructures:

1. **Robust Non-Parametric Framework Verification:** The formal acceptance of H1 and H2 confirms the necessity of non-parametric architectures (Levene's and Mann-Whitney U) in financial analytics, providing a template for mapping tail-risk anomalies without relying on standard Gaussian assumptions.
2. **Quantification of Uniform Capital Flight Cascades:** The mathematical rejection of H3 establishes that emerging market participants execute immediate asset liquidations uniformly upon a shock's disclosure, rendering qualitative severity classifications irrelevant during active sell-offs.
3. **Out-of-Sample Boundary Mapping for Financial AI:** The continuous regression failure boundaries map empirical boundaries demonstrating that parsimonious statistical baselines maintain superior structural integrity over complex ensemble algorithms for level return forecasting in volatile emerging market regimes.

8. LIMITATIONS AND FUTURE WORK

8.1 Current Analytical Limitations

- **Exclusively Lagged Macroeconomic Factors:** While backward-shifting economic variables by one trading day completely immunizes the model from data leakage, fundamental indicators like the CPI are structurally bound to low-frequency monthly institutional releases. This renders daily volatility models partially blind to rapid fundamental shifts during active intra-month trading cycles.
- **Static Binary Shock Formulations:** Systemic crises are currently modeled as distinct, single-day binary entries (`is_shock = 1`). This architecture fails to track the multi-day rolling duration or trailing execution phases of macroeconomic crises (e.g., the prolonged

multi-month economic adjustments of the COVID-19 pandemic or unfolding post-election monetary updates).

- **Absence of Digital Behavioral Tracking Proxies:** Price discovery within highly volatile emerging markets is driven by immediate human psychological reactions. The current data framework relies exclusively on lagging historical macro variables, leaving the model blind to live signals of public anxiety (such as online search trends, mass media stream volume changes, or unstructured digital social sentiment).

8.2 Targeted Future Extensions

- **Integration of Real-Time Digital Behavioral Metrics:** Incorporate daily Google Trends search volume metrics (tracking localized proxy indicators such as "dollar exchange rate" or "inflation rate") alongside scraped social media text streams to construct a composite Digital Panic Index. This framework will evaluate whether real-time behavioral anxiety footprints can detect and explain market regime anomalies faster than low-frequency, backward-looking macroeconomic indicators.
- **Multi-Modal Hybrid Modeling:** Develop nested hybrid validation architectures to statistically contrast traditional fundamental asset pricing models against multi-modal frameworks enriched with digital behavioral metrics, mapping structural improvements in out-of-sample predictive power.
- **Transition to Deep Sequence Architectures:** Replace static machine learning regression tasks with deep recurrent sequence models—such as Long Short-Term Memory (LSTM) networks or temporal transformers—optimized to effectively capture the non-linear, multi-day memory dynamics of financial variance across structural monetary policy regime pivots.